

Benjamin BILLOT

Researcher at Inria

Inria, Epione team
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Experience

2025 - now — **Researcher** — Epione team, Inria Université Côte d'Azur, France
Permanent researcher of the Epione team at Inria Sophia Antipolis, where my work focuses on developing machine learning strategies to robustly analyse medical images.

2022 - 2024 — **Postdoctoral researcher** — MIT, USA
Equivariant and denoising CNNs to decouple spatial and intensity features for motion tracking in fetal MRI
Advisor: Polina Golland

2018 — **Data Scientist** — Founders Factory, London, UK
Developed R-CNN for automated navigation of HTML pages

2016 — **Research assistant** — INSERM, France
Designed a system to study fear-triggering neural mechanisms and electrophysiological responses
Advisors: Thérèse Jay and Marco Pompili

Education

2022 — **PhD in medical image computing** — University College London, UK
Thesis: Bridging generative models and CNNs for domain-agnostic segmentation of brain MRI
Advisor: Juan Eugenio Iglesias

2017 — **MSc in biomedical engineering** — Imperial College London, UK
Thesis: Physics-based generative models in low data regime: segmentation of cortical microscopies.
Advisor: Anil Bharath

2016 — **Diplôme d'ingénieur** — CentraleSupélec, France
Project: Cancellation of respiratory motions for accurate chest CT reconstruction
Advisor: Elisabeth Lahalle

Funded projects and awards

2026 — **ANR JCJC**
Principal coordinator of the SynthUS project (438,000€)

2025 — **IdEx Attractivity Package**
50,000€ awarded by IdEx Université Côte d'Azur

2022 - 2025 — **Reviewer awards**
Best reviewer (MICCAI 2025), Outstanding reviewer (CVPR 2026, MICCAI 2022-2024)

2019 - 2024 — **Short-listed for best paper awards**
MIDL 2019, MIDL 2020, MIDL 2024

2019 — **CMIC platform grant**
£4,000 awarded for a 6-month visit at MIT as a PhD student

Mentoring

2026 - now	**Alvaro Ruiz Lopez – Master student** Domain randomisation for automated quality assessment of brain MRI scans Co-supervised with Ninon Burgos (CNRS)	Inria, France
2026	**Kristina Caleta - PhD student** Segmentation of heart chambers in fetal echocardiography	Inria, France
2023 - now	**Ramya Muthukrishnan - PhD student** Equivariant networks for 3D fetal head pose estimation in motion-aware MRI acquisition at high-resolution Co-supervised with Polina Golland (MIT)	MIT, USA
2025 - now	**Alix de Langlais - PhD student** 3D modelling of proximal humerus fractures for preoperative planning and intraoperative assistance Co-supervised with Hervé Delingette (EPIONE) and Marc-Olivier Gauci (CHU Nice)	Inria, France
2025 - now	**Manasi Kattel - PhD student** Deep learning methods for the analysis and exploitation of ultrasound images CIFRE thesis with KOELIS, co-supervised with Nicholas Ayache (EPIONE) and Hervé Delingette (EPIONE)	Inria, France
2025 - now	**Federica Facente - PhD student** Biomedical models for CT/X-ray image analysis - application to image-guided surgical robotics CIFRE thesis with CARANX MEDICAL, co-supervised with Nicholas Ayache (EPIONE) and Hervé Delingette (EPIONE)	Inria, France
2025 - 2026	**Huiyu Li – Postdoctoral researcher** Robust AI detection tool for lung cancer prognosis Co-supervised with Hervé Delingette (EPIONE) and Charles Marquette (IHU Nice)	Inria, France
2024	**Sebo Diaz - PhD student** Landmark detection for fetal motion tracking in whole-uterus MRI time-series Co-supervised with Elfar Adalsteinsson (MIT) and Polina Golland (MIT)	MIT, USA
2019	**Bo Hyun Song - Master student** Simulation of histological artefacts in medical images Co-supervised with Juan Eugenio Iglesias (UCL)	University College London, UK

Engagement with the scientific community

Event organisation

2026	EURECOM – Inria student days (90 participants)
2023 - 2024	Head of the Organisation committee for the 4th and 5th Boston Medical Imaging Workshops (~100 participants)

Teaching

2018	Introductory Mathematics for computer science (University College London, 10x1h30)

Reviewing

Area Chair	MICCAI (2026), MIDL (2024-2026), IABM (2026)
Reviewer	Journals: Science, Nature Communications, Nature Biomedical Engineering, Medical Image Analysis, IEEE transactions on Medical Imaging, IEEE transactions on Pattern Analysis and Machine Intelligence, Machine Learning for Biomedical Imaging, Scientific Data, NeuroImage, Imaging Neuroscience, Human Brain Mapping, Journal of Neurology Conferences: AAAI (2027), NeurIPS (2026), CVPR (2026), MICCAI (2021-2025), MIDL (2022, 2023), IPMI (2023) Workshop proposals: EurIPS (2025)

Press release

2023 *AI creates high-resolution brain images from low-field strength*
Cynthia E. Keen, PhysicsWorld

Society membership

2020 - now MICCAI member (Medical Image Computing and Computer-Assisted Intervention)
2025 - now IABM member (colloque français d'Intelligence Artificielle en Imagerie Médicale)

Invited presentations

Invited keynote speaker

2026	Improving the robustness of networks for the analysis of clinical images	3IA ANITI, Journée IA pour la santé, France

Invited panellist

2025	Best practices for MICCAI reviews and rebuttals	MICCAI student board webinar, virtual
2024	How to write an award-winning paper?	MIDL Winter Workshop, virtual

Talk: Network conditioning for synergistic learning on partial supervision

2024	Sabuncu Lab seminars	Cornell Tech, New York City, USA
	LEMoN group lab seminars	Harvard Medical School, Boston, USA

Talk: SE(3)-equivariant and noise-invariant 3D rigid motion tracking in brain MRI

2024	ARAMIS seminars	Inria, Paris, France
	IMAGES seminars	Télécom Paris, France
2023	EPIONE seminars	Inria, Nice, France
	MICS seminars	CentraleSupélec, Paris, France

Talk: Robust segmentation of heterogeneous clinical brain MRI scans

2026	Deemea (start-up) seminars	Deemea, Paris, France
2025	Statlearn'25 spring school	Inria, Nice, France
	3IA Côte d'Azur seminars	3IA, Nice, France
2024	Martinos center seminars	Harvard Medical School, Boston, USA
	CMIC-WEISS seminars	UCL, London, UK
2023	LEMoN group lab seminars	Harvard Medical School, Boston, USA

Talk: SynthSeg: domain-randomisation for segmentation of brain MRI of any contrast and resolution

2022	Biomedical imaging and analysis seminars	MIT, Boston, USA
2021	CMIC-WEISS seminars	UCL, London, UK
2020	LCN group seminars	Harvard Medical School, Boston, USA
	Bio-imaging symposium	Imperial College London, UK

Software releases

SynthSeg	Domain-agnostic segmentation of brain scans Python/TensorFlow	GitHub FreeSurfer Matlab	*github.com/BBillot/SynthSeg* *surfer.nmr.mgh.harvard.edu/fswiki/SynthSeg* *mathworks.com/products/matlab.html*
SynthSR	Tool to turn any brain scan into a 1mm T1 scan Python/TensorFlow	GitHub FreeSurfer	*github.com/BBillot/SynthSR* *surfer.nmr.mgh.harvard.edu/fswiki/SynthSR*
Hypo_seg	Segmentation of the hypothalamic subunits Python/TensorFlow	GitHub FreeSurfer	*github.com/BBillot/hypothalamus_seg* *surfer.nmr.mgh.harvard.edu/fswiki/HypothalamicSubunits*
EquiTrack	Motion Tracking in fetal MRI time series Python/PyTorch	GitHub	*github.com/BBillot/EquiTrack*

Summary of publications

3,300+ citations across:

22	Journal articles	4 as first author, 6 as second author
23	Conference articles	7 as first author, 6 as second author

Journal papers

2026 *Toward real-time alignment of 3D CT and 2D X-ray with multi-stage CNNs*
Facente, **Billot**, Gopalakrishnan, Kattel, Wei, Golland, Delingette, Ayache, Berthet-Rayne
Computer Assisted Surgery

Fast segmentation with the NextBrain histological atlas
Puonti, Nolan, Dicamillo, Balbastre, Casamitjana, Mancini, Robinson, Peter, Althonayan, Crampsie, Blackburn, **Billot**, Atzeni, Schmidt, Hughes, Augustinack, Edlow, Zollei, Thomas, Kliemann, Bocchetta, Strand, Holton, Jaunmuktane, Iglesias
Imaging Neuroscience

TumorSynth: integrated brain tumor and tissue segmentation on brain MRI of any resolution and contrast
Wu, **Billot**, Zhao, Li, Liu, Zhuo, Pontillo, Liu, Lin, Zhang, Qiu, Weng, Zhang, Lu, Iglesias, Barkhof, Carrasco
Radiology: Imaging Cancer

Autonomic dysfunction and hypothalamic atrophy in frontotemporal dementia and primary psychiatric disorders
Timar, van Engelen, Venkatraghavan, **Billot**, Fieldhouse, Van't Hooft, de Boer, Koevoets, Hartog, Groot, Oudega, Vijverberg, van der Flier, Barkhof, Pijnenburg
Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring

A case series on longitudinal hypothalamic morphology in mild cognitive impairment
Pagaduan, Marzan, **Billot**
Annals of Geriatric Medicine and Research

2025 *A probabilistic histological atlas of the human brain for MRI segmentation*
Casamitjana, Mancini, Robinson, Peter, Annunziata, Althonayan, Crampsie, Blackburn, **Billot**, Atzeni, Puonti, Balbastre, Schimdt, Hughes, Augustinack, Edlow, Zollei, Thomas, Kliemann, Bochetta, Strand, Holton, Jaunmuktane, Iglesias
Nature

Benchmarking of deep learning methods for generic MRI multi-organ abdominal segmentation
Krishnaswamy, Ciausu, Pieper, Kikinis, **Billot**, Fedorov
Journal of Medical Imaging

A contrast-agnostic method for ultra-high resolution claustrum segmentation
Mauri, Fritz, Mora, **Billot**, Iglesias, Van Leemput, Augustinack, Greve
Human Brain Mapping

Brain MRI signatures across sex and CSF Alzheimer's disease biomarkers
Cheng, He, Gopinath, **Billot**, Iglesias, Wu, Dodge, Wills, Carlyle, Kivisakk, Hyman, Arnold, Das
Brain Communications

2024 *SE(3)-equivariant and noise-invariant 3D rigid motion tracking in brain MRI*
Billot, Dey, Moyer, Hoffmann, Abaci Turk, Gagoski, Grant, Golland
IEEE Transactions on Medical Imaging

Synthetic data in generalizable, learning-based neuroimaging
Gopinath, Hoopes, Alexander, Arnold, Balbastre, **Billot**, Casamitjana, Cheng, Chua, Edlow, Fischl, Gazula, Hoffmann, Keene, Kim, Kimberly, Laguna, Larson, Van Leemput, Puonti, Rodrigues, Rosen, Tregidgo, Varadarajan, Young, Dalca, Iglesias
Imaging Neuroscience

2023 *Robust machine learning segmentation for large-scale analysis of heterogeneous clinical brain MRI datasets*
Billot, Magdamo, Cheng, Arnold, Das, Iglesias
PNAS: Proceedings of the National Academy of Sciences

SynthSeg: segmentation of brain MRI scans of any contrast and resolution without retraining
Billot, Greve, Puonti, Thielscher, Van Leemput, Fischl, Dalca, Iglesias
Medical Image Analysis

SynthSR: a public AI tool to turn heterogeneous clinical brain scans into high-resolution T1 images
Iglesias, **Billot**, Balbastre, Magdamo, Arnold, Das, Edlow, Alexander, Golland, Fischl
Science Advances

Linking brain structure, cognition, and sleep: insights from clinical data
Wei, Ganglberger, Sun, Hadar, Gollub, Pieper, **Billot**, Au, Iglesias, Cash, Kim, Shin, Westover, Thomas
Sleep

Quantitative brain morphometry of portable low-field-strength MRI using super-resolution machine learning
Iglesias, Schleicher, Laguna, **Billot**, Schaefer, McKaig, Goldstein, Sheth, Rosen, Kimberly
Radiology

Brain Growth Charts for Quantitative Analysis of Pediatric Brain MRI Scans with Limited Imaging Pathology
Schabdach, Schmitt, Sotardi, Vossough, Andronikou, Roberts, Huang, Padmanabhan, Ortiz, Gardner, Covitz, Bedford, Mandal, White, Bullmore, Bethlehem, Shinohara, **Billot**, Iglesias, Ghosh, Gur, Satterthwaite, Roalf, Seidlitz, Alexander-Bloch
Radiology

Machine learning of dissection photographs and surface scanning for quantitative 3D neuropathology
Gazula, Tregidgo, **Billot**, Balbastre, William, Herisse, Deden, Casamitjana, Melief, Latimer, Kilgore, Montine, Robinson, Blackburn, Marshall, Connors, Oakley, Frosch, Young, Van Leemput, Dalca, Fischl, Mac Donald, Keene, Hyman, Iglesias
eLife

2022 *SynthMorph: learning contrast-invariant registration without acquired images*
Hoffmann, **Billot**, Greve, Iglesias, Fischl, Dalca
IEEE Transactions on Medical Imaging

In vivo hypothalamic regional volumetry across the frontotemporal dementia spectrum
Shapiro, Todd, **Billot**, Cash, Iglesias, Warren, Rohrer, Bocchetta
NeuroImage Clinical

2021 *Joint super-resolution and synthesis of MPRAGE volumes from clinical MRI with different resolution and contrast*
Iglesias, **Billot**, Balbastre, Tabari, Conklin, Gonzalez, Alexander, Golland, Edlow, Fischl
NeuroImage

A deep learning toolbox for automatic segmentation of subcortical limbic structures from MRI images
Greve, **Billot**, Cordero, Hoopes, Hoffmann, Dalca, Fischl, Iglesias, Augustinack
NeuroImage

2020 *Automated segmentation of the hypothalamus and associated subunits in brain MRI*
Billot, Bocchetta, Todd, Dalca, Rohrer, Iglesias
NeuroImage

Conference papers

2026 *Equivariant Symmetry-Aware Head Pose Estimation for Fetal MRI*
Muthukrishnan, Gagoski, Lee, Grant, Adalsteinsson, **Billot**, Golland
ECCV: European Conference on Computer Vision

LXPose++: an attention-based framework for robust real-time CT to X-ray registration
Facente, Wei, Kattel, Delingette, Ayache, Berthet-Rayne, **Billot**
MICCAI COLAS: workshop on Collaborative Intelligence and Autonomy in Image-guided Surgery

Anatomy-Conditioned Domain Randomization for Zero-Shot 3D Cerebrovascular Segmentation
Pentassuglia, Zhang, **Billot**, Zuluaga
MICCAI: Medical Image Computing and Computer-Assisted Intervention
Early accept

CrossKey: learning cross-modal keypoints for explainable MRI-TRUS rigid prostate registration
Kattel, Billot, Facente, Delingette, Ayache
MICAD: Medical Imaging and Computer-Aided Diagnosis
Best Paper Award

Resource-efficient automatic refinement of segmentations via weak supervision from light feedback
De Langlais, **Billot**, Vidal, Gauci, Delingette
ISBI: International Symposium on Biomedical Imaging

2025 *Learning general-purpose biomedical volume representations using randomized synthesis*
Dey, **Billot**, Wong, Wang, Ren, Grant, Dalca, Golland
ICLR: International Conference on Learning Representations

Spatial regularisation for improved accuracy and interpretability in keypoint-based registration
Billot, Muthukrishnan, Abaci-Turk, Grant, Ayache, Delingette, Golland
MICCAI: Medical Image Computing and Computer-Assisted Intervention

Robust fetal pose estimation across gestational ages via cross-population augmentation
Diaz, **Billot**, Dey, Zhang, Abaci Turk, Grant, Golland, Adalsteinsson
MICCAI: Medical Image Computing and Computer-Assisted Intervention

Fetuses made simple: modelling and tracking of fetal shape and pose
Liu, Wang, Diaz, Abaci Turk, **Billot**, Grant, Golland,
MICCAI: Medical Image Computing and Computer-Assisted Intervention
Early accept

Robust rigid MRI-TRUS registration in prostate cancer using attention-CNN and ICP
Kattel, **Billot**, Facente, Delingette, Ayache
MICCAI ASMUS: Workshop of Advances in Simplifying Medical Ultrasound
Oral presentation

Multi-stage CNN for fast registration of 3D preoperative CTs to 2D intraoperative X-rays
Facente, **Billot**, Gopalakrishnan, Kattel, Wei, Golland, Delingette, Ayache, Berthet-Rayne
MICCAI COLAS: Collaborative Intelligence and Autonomy in Image-guided Surgery

2024 *Network conditioning for synergistic learning on partial annotations*
Billot, Dey, Abaci Turk, Grant, Golland
MIDL: Medical Image with Deep Learning
Short-listed for best paper award, oral presentation

AnyStar: domain randomized universal star-convex 3D instance segmentation
Dey, Abulnaga, **Billot**, Abaci Turk, Grant, Dalca, Golland
WCACV: Winter Conference on Applications of Computer Vision
Early accept

Quantifying white matter hyperintensity and brain volumes in heterogeneous clinical and low-field portable MRI
Laso, Cerri, Sorby, Guo, Mateen, Goebl, Wu, Liu, Li, Young, **Billot**, Puonti, Sze, Sheth, Rosen, Kirsch, Strisciuglio, Wolterink, Eshaghi, Barkhof, Kimberly, Iglesias
ISBI: International Symposium on Biomedical Imaging

Association of brain MRI signatures with Alzheimer disease CSF biomarkers
Cheng, Gopinath, **Billot**, Iglesias, Touroutoglou, Kivisäkk, Wu, Dodge, Hyman, Arnold, Das
Alzheimer's Association International Conference

2023 *Domain-agnostic segmentation of thalamic nuclei from joint structural and diffusion MRI*
Tregidgo, Soskic, Olchanyi, Althonayan, **Billot**, Maffei, Golland, Yendiki, Alexander, Bocchetta, Rohrer, Iglesias
MICCAI: Medical Image Computing and Computer-Assisted Intervention
Early accept

Equivariant and denoising CNNs to decouple intensity and spatial features for motion tracking in fetal brain MRI
Billot, Moyer, Karani, Hoffmann, Abaci Turk, Grant, Golland
MIDL: Medical Image with Deep Learning, short paper track

2022 *Robust segmentation of brain MRI in the wild with hierarchical CNNs and no retraining*
Billot, Magdamo, Arnold, Das, Iglesias
MICCAI: Medical Image Computing and Computer-Assisted Intervention

Super-resolution of portable low-filed MRI in real scenarios: integration with denoising and domain adaptation
Laguna, Schleicher, **Billot**, Schaefer, McKaig, Goldstein, Sheth, Rosen, Kimberly, Iglesias
MIDL: Medical Image with Deep Learning, short paper track

2021 *Joint segmentation of multiple sclerosis lesions and brain anatomy in MRI scans of any contrast and resolution*
Billot, Cerri, Van Leemput, Dalca, Iglesias
ISBI: International Symposium on Biomedical Imaging
Oral presentation

Learning MRI contrast-agnostic registration
Hoffmann, **Billot**, Iglesias, Fischl, Dalca
ISBI: International Symposium on Biomedical Imaging

2020 *Partial volume segmentation of brain MRI scans of any contrast and resolution*
Billot, Robinson, Dalca, Iglesias
MICCAI: Medical Image Computing and Computer-Assisted Intervention
Oral presentation, early accept

A learning strategy for contrast-agnostic MRI segmentation
Billot, Greve, Van Leemput, Fischl, Dalca, Iglesias
MIDL: Medical Image with Deep Learning
Short-listed for best paper award, oral presentation

2019 *Image synthesis with a convolutional capsule generative adversarial network*
Bass, Dai, **Billot**, Arulkumaran, Creswell, Clopath, De Paola, Bharath
MIDL: Medical Image with Deep Learning
Short-listed for best paper award, oral presentation

Deep reinforcement learning for subpixel neural tracking
Dai, Dubois, Arulkumaran, Campbell, Bass, **Billot**, Uslu, De Paola, Clopath, Bharath
MIDL: Medical Image with Deep Learning

Conference abstracts

2026 *Resource-efficient automatic refinement of segmentations via weak supervision from light feedback*
De Langlais, **Billot**, Vidal, Gauci, Delingette
Colloque Français d'Intelligence Artificielle en Imagerie Biomédicale

Robust rigid MRI-TRUS registration in prostate cancer using attention-CNN and ICP
Kattel, **Billot**, Facente, Delingette, Ayache
Colloque Français d'Intelligence Artificielle en Imagerie Biomédicale

2025 *Network conditioning for synergistic learning on partial annotations*
Billot, Dey, Abaci Turk, Grant, Golland
Colloque Français d'Intelligence Artificielle en Imagerie Biomédicale

2017 *Physics-based models in low data regime: application to segmentation of cortical microscopies*
Billot, Bass, Bharath
Human brain Project 4th summer school